Adaptive Opponent Modeling for Adversarial Co-Training in MARL

A controlled predator–prey study: inferring a varying opponent's latent strategy and planning against it

Rajdeep Singh
REL Lab, University of Southern California

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Abstract

In adversarial co-training each team faces an opponent whose strategy varies and adapts; a best response that overfits to one opponent loses robustness to strategies it has not recently seen. The remedy we pursue is an *uncertainty-aware* model of the opponent's latent strategy (Part 1), *sampled inside a model-based planner* (Part 2). This note instantiates and validates both parts in a controlled predator–prey game. The opponent—the prey, team red—draws a hidden strategy each episode; the controlled team—the predators, team blue—cannot observe it and must infer it from red's behavior, then plan. We instantiate the latent strategy z as a discrete intent so inference can be scored against ground truth. We find: **(i)** the opponent's latent strategy is decisive—a blue policy that observes z does +51%; **(ii)** it is inferable from red's trajectory with calibrated uncertainty—an encoder $q_\phi(z \mid \tau^{\text{red}})$ recovers it with accuracy $0.37 \to 0.97$ over $3 \to 25$ observed steps as its posterior entropy collapses; **(iii)** a *point-estimate* opponent model is brittle—fed a wrong guess it loses 47%, exactly the failure uncertainty-aware modeling is meant to prevent; and **(iv)** sampling the uncertainty-aware opponent model inside a planner yields the most robust best response, +61% over an opponent-blind baseline and *above an oracle* blue policy told the true strategy. An ablation that flattens the belief shows the inference, not the lookahead, is the active ingredient (+40%). The study validates the proposal's Part 1/Part 2 loop on its first target domain and motivates the full variational-encoder, CPL, and MuZero-style instantiation.

1 Background: adaptive opponent modeling for co-training

Adversarial co-training—training both teams together rather than scripting one—is necessary for robust strategies in complex games, but it is hard: each team faces a *non-stationary* opponent, policies can fall into cyclic (rock–paper–scissors) dynamics, and a team readily *overfits to the current opponent*, losing to strategies that have not appeared recently. Existing approaches underserve exactly the opponent that varies at test time: league play is brute force and brittle to novel opponents; CTDE methods (e.g. MADDPG) assume centralized access to the opponent and model *no* uncertainty in its strategy; model-based opponent modeling assumes known goals or clones behavior to a moving target.

The approach. Model the opponent's strategy with *calibrated uncertainty* and *plan* against that model. Two coupled parts:

Part 1 (uncertainty-aware opponent model).
A variational encoder $q_\phi(z \mid \tau^{\text{red}})$ maps the opponent's (red's) trajectory to a latent strategy z with a posterior that expresses how sure we are; a latent-conditioned opponent policy $\pi^{\text{red}}(a \mid s, z)$ reconstructs red's behavior (by behavior cloning or contrastive preference learning). The model must stay robust to opponents that adapt and use varying strategies.

Part 2 (plan against it).
The controlled team (blue) plans with a model-based search that *samples* the opponent's moves from $\pi^{\text{red}}(\cdot \mid s, z)$ and conditions its value $v^{\text{blue}}(s, z)$ on the inferred strategy, so the plan is aware of which opponent it currently faces.

This study. We validate the loop in the proposal's first domain, predator–prey, in a form where every quantity can be measured. The opponent (prey, red) is assigned a hidden *intent* each episode—one of four corners it is rewarded for haunting—which the controlled predators (blue) cannot observe. The intent is the latent strategy z, made discrete so the encoder's belief can be checked against truth and so "the opponent uses varying strategies" has an exact meaning: a fresh z every episode. The environment is otherwise fixed, so z is *not* readable from a single observation and can only be had by modeling red's behavior over time.

2 Setup

Three blue predators P pursue one red prey q in a fixed 2D arena (JaxMARL MPE, five discrete actions, 25-step episodes, two fixed obstacles). At reset the prey draws $z \sim \text{Uniform}\{0, 1, 2, 3\}$ indexing a corner $c_z \in \{\pm 0.8\}^2$. With $k_i^t = \mathbf{1}[\|x_i^t - x_q^t\| < \rho_i + \rho_q]$ marking predator i in contact with the prey,

$$r^{\text{blue},t} = \lambda_{\text{tag}} \sum_{i \in P} k_i^t, \qquad r^{\text{red},t} = -\lambda_{\text{tag}} \sum_{i \in P} k_i^t + \lambda_{\text{int}} \mathbf{1}\big[\|x_q^t - c_z\| < r_{\text{int}}\big], \tag{1}$$

λ_{tag}=10, λ_{int}=4, r_{int}=0.35: the prey loiters at its assigned corner when safe while evading. The prey observes its own z; *the predators do not.* Each policy is a MAPPO co-training run (centralized critic, 3 seeds); we report captures per episode, cap $= \lambda_{\text{tag}}^{-1} \sum_t r^{\text{blue},t}$, over 300 episodes.

3 Method (the controlled instantiation)

Part 1 — an uncertainty-aware model of the opponent's strategy. From the prey's first k positions $\tau_{0:k}^{\text{red}}$ we fit $q_\phi(z \mid \tau_{0:k}^{\text{red}})$, here a classifier whose softmax is a posterior belief $\mathbf{b}_k$ over the four latent strategies. The belief, not a point estimate, is the object we carry: it sharpens online as more of the opponent is observed, which is the discrete instance of the proposal's variational posterior over z. The opponent policy model is $\pi^{\text{red}}(a \mid s, z)$, the prey's own behavior under strategy z (the target a latent-conditioned BC/CPL model is trained to reproduce).

Part 2 — planning against the uncertainty-aware model. At state s_t with belief $\mathbf{b}_t$, blue scores each candidate joint action u by a short rollout in which red's moves are *sampled from the* *opponent model under a strategy drawn from the belief*:

$$Q(s_t, u) = \mathbb{E}_{z \sim \mathbf{b}_t} \left[\sum_{h=0}^{H-1} \gamma^h r_{t+h}^{\text{blue}} \; - \; w \min_{i \in P} \| x_i^{t+H} - x_q^{t+H} \| \;\middle|\; u_t{=}u, \; a_{t+h}^q \sim \pi^{\text{red}}(\cdot \mid s_{t+h}, z) \right], \qquad (2)$$

and acts $u_t = \arg\max_u Q(s_t, u)$ (H=5, γ=0.95, K=8 rollouts, strategies shared across candidates). Marginalizing over $\mathbf{b}_t$ makes the plan *uncertainty-aware*: while the belief is flat blue hedges, and as it sharpens blue commits. The leaf term rewards ending near the imagined future prey, which under a sampled strategy sits at the believed corner—an interception. This is the simulator-based instance of the proposal's tree search that samples $\pi^{\text{red}}(\cdot \mid z)$ at each node and scores with a strategy-aware value $v^{\text{blue}}(s, z)$.

4 Results

(i) The opponent's latent strategy is inferable (Table 1, Figure 1). The prey's motion clusters by strategy (Figure 1, left), and the encoder recovers z with accuracy 0.37→0.97 over 3→25 steps as its posterior entropy collapses from 1.35 to 0.03 nats—a usable, increasingly-confident belief well before the episode ends.

Table 1: Recovery of the opponent's latent strategy from its first k steps (4-way; chance 0.25).

steps k	3	5	8	12	18	25
accuracy	0.37	0.70	0.74	0.84	0.93	0.97
posterior entropy (nats)	1.35	0.75	0.52	0.31	0.09	0.03

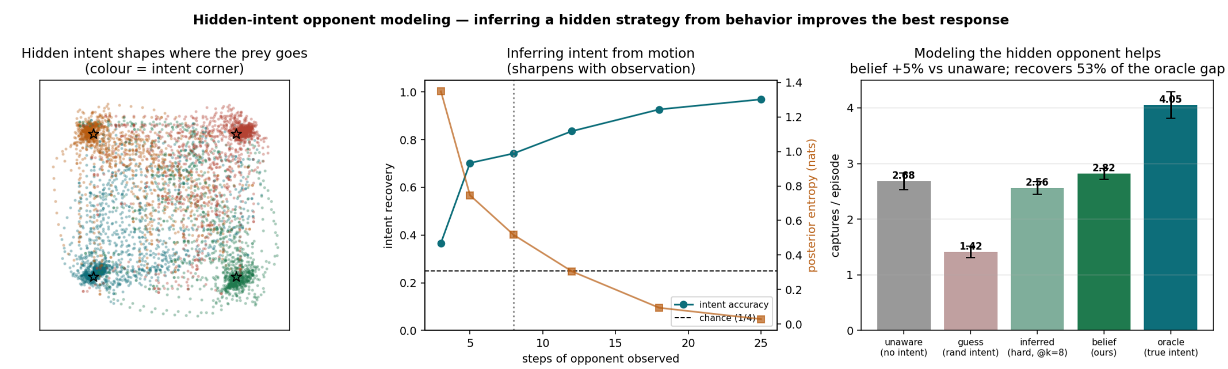

Figure 1: **Left:** red occupancy coloured by its hidden latent strategy—four clusters in one fixed arena. **Middle:** the encoder's accuracy rises and posterior entropy falls with steps of the opponent observed. **Right:** blue captures per episode across the opponent-model signals it is given.

(ii) The strategy is decisive, and a point estimate is brittle. A blue policy that observes the true z catches the prey 4.05 times/episode vs. 2.68 for an opponent-blind baseline (+51%): knowing which opponent one faces matters a great deal. But a blue policy trained on *perfect* z and fed a *wrong* one collapses to 1.42 (−47%)—the precise brittleness of opponent models that ignore their own uncertainty, which the uncertainty-aware belief exists to prevent.

(iii) Planning against the uncertainty-aware model is the most robust response (Table 2, Figure 2). The planner of Eq. (2) reaches **4.31** captures: +61% over the opponent-blind baseline, +53% over a reactive policy given the same belief, and *above the oracle reactive policy* (4.05)—planning interceptions against the inferred opponent beats reacting even with the true strategy in hand. The control isolates the cause: a planner given a *flat* belief (lookahead, no opponent inference) reaches only 3.07, so the inferred opponent model supplies +40%. The opponent inference, not the lookahead, is what wins.

Table 2: Blue captures per episode against the varying red opponent (mean ± std, 3 seeds × 300 episodes).

blue predator	captures / ep	vs. opponent-blind
opponent-blind (no strategy info)	2.68 ± 0.15	—
reactive, inferred belief	2.82 ± 0.10	+5%
planner, flat belief (ablation)	3.07 ± 0.17	+15%
oracle, true strategy (reactive)	4.05 ± 0.24	+51%
planner, inferred belief (Part 1+2)	$\mathbf{4.31 \pm 0.36}$	**+61%**

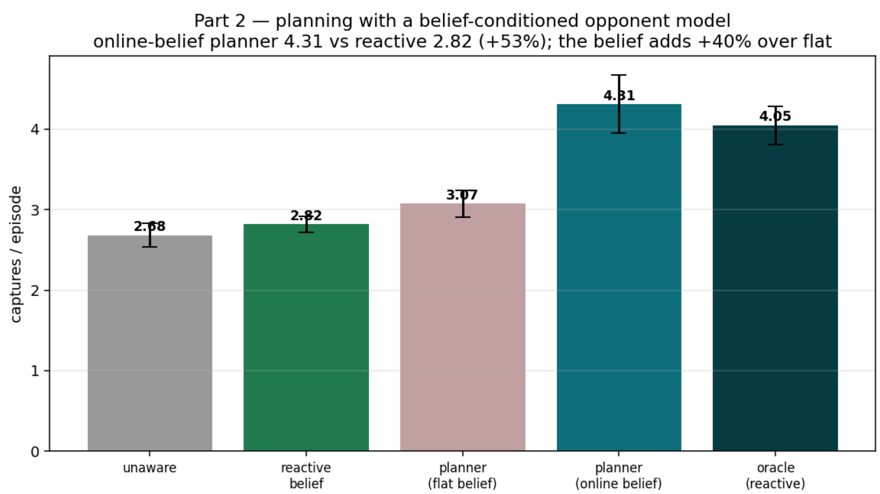

Figure 2: Planning against the uncertainty-aware opponent model (online belief) tops every reactive policy, including the oracle. Ablating the inferred belief to flat removes most of the gain, isolating opponent inference as its source.

5 Part 1 without labels: predict, don't reconstruct

The encoder above used intent labels. A Part 1 that scales should recover the opponent's strategy *self-supervised*, from trajectories alone—and the *objective* matters. We compare two self-supervised encoders of the prey trajectory on identical data, both bottlenecked to a 2-D latent (so the latent *is* the figure) and scored only afterward against the four ground-truth intents:

- **VAE (generative):** encode the first 12 steps, then *reconstruct* them (ELBO). It must model every detail—the random start and the evasion noise included.

- **JEPA (predictive):** encode the first 12 steps, then *predict the representation* of the future window (steps 12–24, the corner approach) through an EMA target encoder—no reconstruction. It keeps only the predictable structure (where the prey is heading = its strategy) and discards the noise.

Table 3: Unsupervised recovery of the opponent's strategy from a 2-D latent (4-way; probe chance 0.25). The predictive encoder wins on both metrics and beats a supervised classifier trained on the raw window.

self-supervised encoder	probe acc.	GMM ARI
VAE (generative, reconstruct)	0.54	0.14
JEPA (predictive, predict repr.)	**0.88**	**0.67**
supervised classifier (raw window)	0.85	—

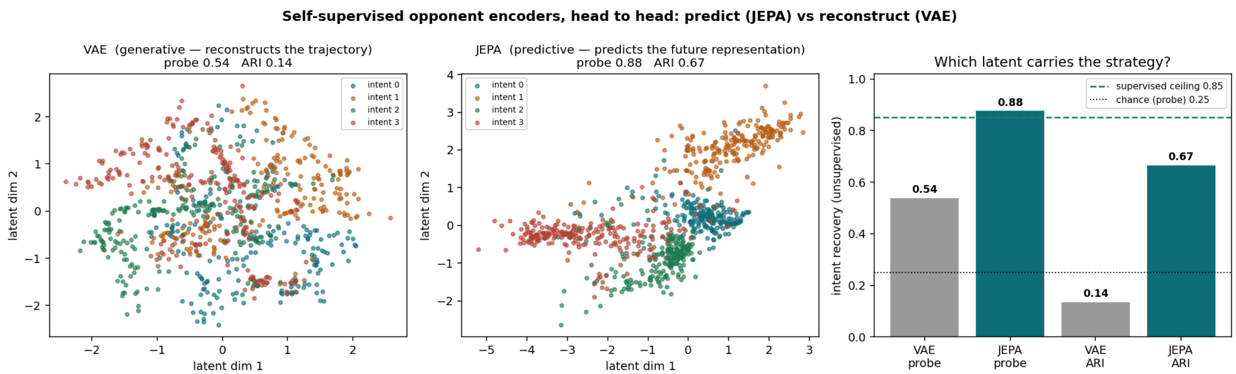

Figure 3: The generative VAE latent blends the four intents (probe 0.54); the predictive JEPA latent separates them (probe 0.88, ARI 0.67), above the supervised ceiling. Same data, same capacity, same 2-D bottleneck—only the objective differs.

The predictive objective recovers the opponent's strategy where reconstruction cannot (Table 3, Figure 3)—and, notably, its smooth 2-D latent is *more* linearly separable than the raw window a supervised classifier sees. This is LeCun's JEPA principle (predict in representation space, not input space; cf. V-JEPA 2) applied to opponent modeling: the part of an opponent's behavior worth encoding is the part you can *predict*, and a generative encoder squanders capacity on the rest. It is also the cleaner Part 1 for the proposal: no labels, and a latent that drops straight into the belief and the planner.

6 Relation to the proposal and next steps

This study validates the proposal's loop end to end on its first domain. Part 1: the opponent's latent strategy is recoverable from its trajectory into a calibrated belief, and—crucially for co-training—a model that ignores that uncertainty is brittle to a strategy it did not expect (−47%), the overfitting failure adaptive opponent modeling targets. Part 2: sampling the uncertainty-aware opponent model inside a planner produces the most robust best response against an opponent that varies its strategy every episode, beating even an oracle that is handed the answer. The natural next steps are exactly the full proposal: replace the discrete classifier with the variational encoder $q_\phi(z \mid \tau^{\mathrm{red}})$ and a latent-conditioned π_ψ^{red} (behavior cloning, then contrastive preference learning with planner-generated counterfactuals), and the simulator rollout with a MuZero-style learned model and a strategy-conditioned value $v^{\mathrm{blue}}(s, z)$, then close the co-training loop so red and blue adapt to each other.